Adversarially Trained Baseline for Sift

A GAN-style alignment scorer on frozen (intent, hunk) embeddings, compared with the cross-encoder and a no-training cosine baseline

PROJECT	DELIVERABLE	TEAM NO	TEAM MEMBER	INTERPRETATION
Sift	**Deliverable 4: GAN Results**	**Team 24 (single member)**	**Kundhave S, CB.SC.U4CSE23037**	**Adversarial baseline on frozen embeddings**

1. Executive Summary and Motivation

Interpretation. Sift has no GAN in its design, so this deliverable is implemented as an **adversarially trained baseline** that works on frozen sentence embeddings of the (intent, hunk) pairs. A generator learns to produce synthetic aligned-hunk embeddings from noise and the intent embedding; a discriminator learns to separate real aligned pairs from generated ones and is then used as an alignment scorer. The point is a comparison: does adversarial training on frozen embeddings get close to the fine-tuned cross-encoder?

- On the test split the unsupervised variant reaches macro-F1 0.630, NDCG@5 0.864 and pairwise AUC 0.784 (random tier) / 0.708 (hard tier); the cross-encoder reaches 0.680, 0.868, 0.800 / 0.726.
- The hybrid variant (misaligned training pairs also fed to D as negatives) reaches 0.625, 0.871, 0.807 / 0.705.
- Neither GAN variant beats the training-free frozen cosine dissimilarity on NDCG@5 (0.874) or random-tier AUC (0.814); on macro-F1 they are +0.016 and +0.010 above it. The adversarial training therefore adds little over a cosine score.
- Training was unstable: the generator was collapsed (mean pairwise cosine above 0.9) for 9 epoch(s) of 40 in the unsupervised run and 4 in the hybrid run, and validation AUC peaked at epoch 11 then declined.

2. Architecture Formulations

Notation: e_i and e_h are unit-norm 384-dimensional embeddings of the consolidated intent and of the hunk text (frozen all-MiniLM-L6-v2, max length 512); $z \sim N(0, I_{64})$.

Generator G (a 2-hidden-layer MLP with 512 units and LeakyReLU 0.2, output L2-normalised) maps noise and intent to a synthetic aligned-hunk embedding:

$$\tilde{e}_h = G(z, e_i) = \frac{\mathrm{MLP}([e_i; z])}{\|\mathrm{MLP}([e_i; z])\|_2}$$

Discriminator D scores a pair from the features $[e_i; e_h; e_i * e_h; |e_i - e_h|]$ (MLP 1536-512-256-1, dropout 0.2) and is trained with the standard non-saturating GAN losses, using only *aligned* training pairs as real:

$$\mathcal{L}_D = -\mathbb{E}_{(e_i, e_h) \sim \text{aligned}} \log \sigma(D(e_i, e_h)) - \mathbb{E} \log\left(1 - \sigma(D(e_i, G(z, e_i)))\right)$$

$$\mathcal{L}_G = -\mathbb{E} \log \sigma(D(e_i, G(z, e_i)))$$

Hybrid variant adds real misaligned pairs as extra negatives for D (a supervised signal that the pure GAN objective does not use):

$$\mathcal{L}_D^{\text{hyb}} = \mathcal{L}_D - \mathbb{E}_{(e_i, e_h) \sim \text{misaligned}} \log\left(1 - \sigma(D(e_i, e_h))\right)$$

Scoring. The misalignment score of a pair is $s = -D(e_i, e_h)$ (low realism = likely misaligned); the generator is discarded at inference. A threshold is tuned on validation macro-F1 after a monotone squashing of s, so the GAN scores are not calibrated probabilities. **Collapse monitor.** The mean pairwise cosine of 256 generated embeddings; 1.0 means every generated embedding is identical (complete mode collapse).

$$\text{div} = \frac{1}{n(n-1)} \sum_{a \neq b} \cos(\tilde{e}^{(a)}, \tilde{e}^{(b)})$$

Reference baseline. Frozen cosine dissimilarity, $s = -\cos(e_i, e_h)$, with no trainable parameter; it shows how much the adversarial machinery adds over the embeddings alone.

3. Experimental Setup and Training Protocol

Item	Setting
Data	Sift-IA v1.0 train / val / test (repository-disjoint). Real = aligned training pairs (observed label, so label noise is present); val for checkpoint and threshold; test for reporting.
Embedder	sentence-transformers/all-MiniLM-L6-v2 (frozen), 90.9 MB, CPU
Optimiser	Adam, lr 2e-4, betas (0.5, 0.999), batch 128, 40 epochs, seed 473, CPU only
Checkpoint	best validation AUC (epoch 11 unsupervised, epoch 3 hybrid; best val AUC 0.729 and 0.733)
Parameters	D 918,529; G 689,536; D size 3.68 MB
Training time	132 s (unsupervised), 143 s (hybrid), on top of a one-off embedding pass
Evaluation	Same test split, label_clean, same metrics and tiers as the cross-encoder; threshold from validation

4. Quantitative Comparison

Model	Macro-F1	AUC random	AUC hard	NDCG@5	ECE	Size MB	CPU p50 ms / pair	Trainable
Main: ModernBERT cross-encoder	0.680	0.800	0.726	0.868	0.066	596.1	94.5	149.0 M
INT8 ONNX student (deployed)	0.667	0.784	0.681	0.869	0.051	22.9	8.5	22 M
Frozen cosine dissimilarity	0.614	0.814	0.719	0.874	0.277	90.9	28.1	0 trainable
GAN-style, unsupervised D	0.630	0.784	0.708	0.864	0.264	94.5	28.4	D 0.92 M + G 0.69 M
GAN-style, hybrid D	0.625	0.807	0.705	0.871	0.269	94.5	28.2	D 0.92 M + G 0.69 M

Notes: CPU latency of the GAN-style and cosine rows is the frozen embedder on one pair (p50 28.1 ms, p95 64.0 ms) plus, for the GAN rows, the discriminator (p50 0.27 ms); the cross-encoder figure is fp32 PyTorch on an idle machine and the INT8 figure was taken while a training job shared the CPU, so they are not strictly comparable. ECE of the GAN and cosine rows is large because their scores are squashed, not calibrated, probabilities. Size for the GAN rows is the embedder plus D (G is not needed at inference).

Confusion counts on the 2,700 test hunks at the validation-tuned threshold (the thresholds of the GAN rows live on a squashed score scale and are not comparable with the cross-encoder's probability threshold):

Scorer	Threshold	TN	FP	FN	TP	Precision	Recall	Specificity
Main cross-encoder	0.30	1841	265	311	283	0.516	0.476	0.874
Frozen cosine	0.54	1802	304	378	216	0.415	0.364	0.856
GAN-style, unsupervised D	0.84	1713	393	321	273	0.410	0.460	0.813
GAN-style, hybrid D	0.72	1778	328	356	238	0.420	0.401	0.844

5. Training Curves and Instability Analysis

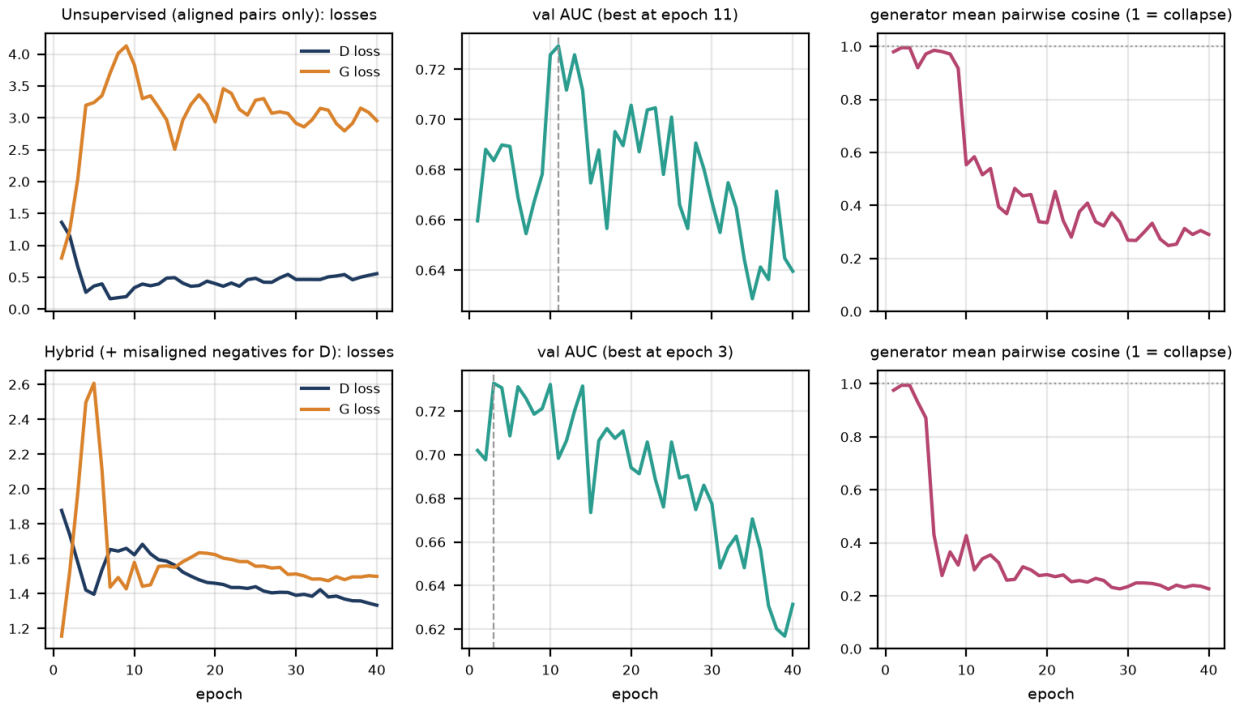


Figure 1. Training history of both variants (experiments/results/gan.json): losses, validation AUC and generator diversity per epoch.

- **Mode collapse at the start.** The generator output is almost constant at epoch 1 (mean pairwise cosine 0.98 unsupervised, 0.98 hybrid); it diversifies later (minimum 0.25 and 0.22, final 0.29 and 0.23). In the unsupervised run diversity stays above 0.9 for 9 of 40 epochs (last such epoch: 9); in the hybrid run for 4 (last: 4).
- **Discriminator dominance.** The lowest D loss (0.16 at epoch 7) coincides with a generator loss of 3.70 and a generator diversity of 0.98, consistent with D easily rejecting near-identical generated embeddings; later D loss settles near 0.56 and G loss near 2.95.
- **Overfitting of the scorer.** Validation AUC peaks at 0.729 (epoch 11) and ends at 0.640; the hybrid variant peaks at 0.733 (epoch 3) and ends at 0.631. Best-by-validation checkpointing was therefore needed and means the validation set influenced the model choice.
- **Run summary.** Unsupervised: D loss 1.36 -> 0.56, G loss 0.80 -> 2.95, epoch time about 3.2 s. Hybrid: D loss 1.88 -> 1.33, G loss 1.16 -> 1.50. The hybrid D loss stays higher because it includes the misaligned-negative term.
- **No convergence guarantee.** A single seed and a fixed 40 epochs were run; GAN training is seed-sensitive and different seeds could give different collapse patterns. This was not tested.

6. Failure Analysis

- **One-class training signal.** The unsupervised D sees only aligned pairs, so it learns what real pairs look like, not what misaligned pairs look like; its hard-tier AUC (0.708) is 0.018 below the cross-encoder's and its macro-F1 is 0.050 lower.
- **Adversarial training adds little over the embedding.** The no-training cosine baseline already gives NDCG@5 0.874 and random-tier AUC 0.814, above both GAN variants on those two metrics. The hybrid variant gains +0.023 random-tier AUC over the unsupervised one, but that comes from the supervised negatives, not from the adversarial game.
- **Where a frozen baseline does well and why that matters.** Frozen cosine matches or beats the cross-encoder on NDCG@5 (0.874 vs 0.868) and random-tier AUC (0.814 vs 0.800) with no training. On the hard tier the cross-encoder is only marginally ahead (0.726 vs 0.719), so frozen embeddings already carry much of the signal these tiers measure; the hybrid GAN variant also edges the cross-encoder on random-tier AUC (0.807 vs 0.800). The cross-encoder's advantage is mainly macro-F1 (0.680 vs 0.614) and hard-tier AUC over the weakest scorers, not a clear win in ranking.
- **Poor calibration and unstable thresholds.** The GAN scores have ECE 0.26 / 0.27; a threshold tuned on validation (macro-F1 0.630, recall 0.460, precision 0.410 for the unsupervised variant) cannot be read as a probability.

7. Visual Comparison

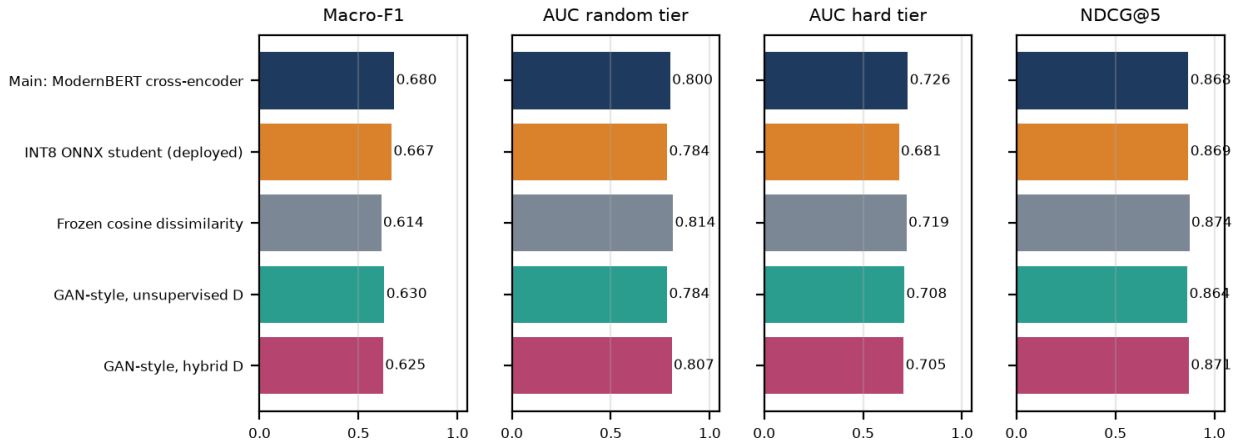


Figure 2. Test-split comparison of the five scorers (same split, same thresholding rule).

8. Scope, Limitations and Conclusion

The adversarially trained baseline is a working but weak alignment scorer: it reaches macro-F1 0.630 / 0.625 against 0.680 for the cross-encoder, is no better than a cosine score on ranking, shows early mode collapse and later overfitting, and is poorly calibrated. Its advantages are size of the trainable part (a 3.7 MB discriminator) and no fine-tuning of the encoder, but the frozen embedder it depends on is itself 91 MB. It is not part of the shipped Sift pipeline. This deliverable follows the interpretation stated in section 1.

- Labels in Sift-IA v1.0 are construction-determined (a misaligned hunk is a donor hunk from a different commit); there is no human validation.
- Misaligned hunks are constructed, so a model could learn construction artefacts; the hard tier mitigates this but does not remove it.
- Latency and size come only from the machine they were measured on (Intel i7-13650HX CPU, RTX 4050 Laptop GPU with 6 GB); every run is a single seed (473), no confidence intervals.
- GAN rows are one 40-epoch run per variant on CPU; no repeat seeds, no hyper-parameter search. The comparison with the cross-encoder is between very different training budgets.
- Command to regenerate the GAN results on another machine: `PYTHONPATH=src python experiments/run_gan.py`